

Nutrition Interventions in Low- and Middle-Income Countries: Systematic Evidence Review

Based on 100 top-ranked papers from a multi-source systematic search (PubMed + OpenAlex) 57 papers with structured full text (PMC); 43 abstract-only Date of synthesis: 26 May 2026

Table of Contents

- 1. Summary Table: Intervention Rankings
- 2. Tier 1 Interventions
- 3. Tier 2 Interventions
- 4. Tier 3 Interventions
- 5. Cross-Cutting Findings
- 6. Methodology Notes

1. Summary Table: Intervention Rankings

Rank	Intervention	Type	Evidence	Cost-Effectiveness	Scalability	Primary Target
1	Vitamin A supplementation (children 6–59 mo)	Nutrition-specific	A	Very High	Proven national	Under-5 mortality, morbidity
2	Iron-folic acid supplementation (pregnancy)	Nutrition-specific	A	Very High	Proven national	Maternal anaemia, LBW
3	Multiple micronutrient supplementation (pregnancy)	Nutrition-specific	A	Very High	Proven national	LBW, SGA, stillbirth
4	Large-scale food fortification (iron, iodine, folic acid)	Nutrition-specific	A	Very High	Proven national	Anaemia, IDD, NTDs
5	Exclusive breastfeeding promotion	Nutrition-specific	A	Very High	Proven national	Under-5 mortality, infections
6	Complementary feeding education ± food provision	Nutrition-specific	A	High	Proven subnational	Stunting, wasting (6–24 mo)

Rank	Intervention	Type	Evidence	Cost-Effectiveness	Scalability	Primary Target
7	Micronutrient powders (home fortification)	Nutrition-specific	A	High	Proven national	Anaemia, iron deficiency (6–23 mo)
8	Zinc supplementation (preventive, children)	Nutrition-specific	A	High	Proven subnational	Diarrhoea morbidity
9	Zinc for diarrhoea treatment	Nutrition-specific	A	Very High	Proven national	Diarrhoea duration
10	Management of SAM (RUTF-based, community)	Nutrition-specific	A	High	Proven subnational	SAM recovery, mortality
11	Lipid-based nutrient supplements (children 6–23 mo)	Nutrition-specific	A	Moderate	Growing	Stunting, wasting, underweight
12	Prenatal small-quantity LNS	Nutrition-specific	B+	Moderate	Growing	Birth size, newborn stunting
13	Management of MAM (RUSF/LNS)	Nutrition-specific	B+	Moderate	Proven subnational	MAM recovery
14	Iron supplementation (school-age children/adolescents)	Nutrition-specific	B+	High	Proven subnational	Anaemia, cognition
15	WASH interventions (combined)	Nutrition-sensitive	B	Moderate	Proven subnational	HAZ, diarrhoea reduction
16	Conditional cash transfers	Nutrition-sensitive	B	Moderate	Proven national	Stunting, health service use
17	Community case management of pneumonia	Nutrition-specific	B+	High	Proven subnational	Pneumonia mortality
18	Balanced protein-energy supplementation (pregnancy)	Nutrition-specific	B	Moderate	Growing	Birthweight
19	Nutrition education during pregnancy	Nutrition-specific	B+	High	Proven subnational	IFAS compliance, anaemia
20	IMCI strategy	Nutrition-sensitive	B	Moderate	Proven subnational	Under-5 mortality
21	Calcium supplementation (pregnancy)	Nutrition-specific	B	Moderate	Growing	Pre-eclampsia

Rank	Intervention	Type	Evidence	Cost-Effectiveness	Scalability	Primary Target
22	Vitamin D supplementation (pregnancy)	Nutrition-specific	C+	Low–Moderate	Requires investment	LBW, preterm birth
23	Agricultural interventions (home gardens, biofortification)	Nutrition-sensitive	C	Low	Growing	Dietary diversity
24	Growth monitoring and promotion	Nutrition-sensitive	C	Moderate	Proven subnational	Nutritional status detection

2. Tier 1 Interventions

Strong evidence (A) + proven scalability + high/very high cost-effectiveness

2.1 Vitamin A Supplementation (Children 6–59 Months)

Evidence Rating: A

The most extensively studied child survival intervention in the database, supported by multiple Cochrane reviews and large meta-analyses spanning over 1.2 million children.

Key Effect Sizes:

Outcome	Effect	Source
All-cause mortality (fixed-effect)	RR 0.88 (95% CI: 0.83–0.93); 19 trials, n=1,202,382	PMID 35294044 (Cochrane 2022), high-certainty
All-cause mortality (random-effects)	RR 0.76 (95% CI: 0.66–0.88)	PMID 28282701 (Cochrane 2017)
All-cause mortality (6–59 mo)	RR 0.75 (95% CI: 0.64–0.88); 9 studies	PMID 21501438 (2011), fulltext
Diarrhoea-specific mortality	RR 0.88 (95% CI: 0.79–0.98); 9 trials, high-certainty	PMID 35294044
Diarrhoea incidence	RR 0.85 (95% CI: 0.82–0.87); 15 trials	PMID 35294044
Measles incidence	RR 0.50 (95% CI: 0.37–0.67); 6 studies	PMID 28282701
Night blindness	RR 0.32 (95% CI: 0.21–0.50)	PMID 35294044
Xerophthalmia	RR 0.31 (95% CI: 0.22–0.45)	PMID 21868478, fulltext
Vitamin A deficiency	RR 0.71 (95% CI: 0.65–0.78); high-certainty	PMID 35294044

Subgroup Findings (from full-text papers): - Asia subgroup shows larger mortality reduction: RR 0.69 (95% CI: 0.61–0.79) vs. Africa RR 0.85 (95% CI: 0.73–0.98) [PMID 21868478] - Effect attenuated in recent large trials: well-concealed trials only RR 0.91 (95% CI: 0.85–0.97); trials since 2000 RR 0.96 (95% CI: 0.89–1.03) [PMID 35584136, fulltext] - Nepal-specific modelling: only 3 fewer deaths per 1,000 children supplemented in current context [PMID 35584136]

Important Caveats: - Neonatal VAS shows **no overall benefit**: RR 0.97 (95% CI: 0.89–1.06) at 6 months across 11 trials, n=163,567 [PMID 30425075, fulltext IPD]. Benefit confined to Asia (RR 0.87, 95% CI: 0.77–0.98) and populations with high maternal VAD prevalence (RR 0.87, 95% CI: 0.80–0.94). Possible harm in Africa (RR 1.06, 95% CI: 0.98–1.15). - VAS in infants <6 months: no benefit (RR 1.05, 95% CI: 0.88–1.26) and increased bulging fontanelle risk (RR 1.55, 95% CI: 1.05–2.28) [PMID 21975770] - Adverse effect: vomiting within 48 hours (RR 1.97, 95% CI: 1.44–2.69) [PMID 35294044]

Mechanism: Vitamin A maintains epithelial barrier integrity, supports immune cell differentiation, and enhances mucosal defence against enteric and respiratory pathogens.

Cost-Effectiveness: Very High. One of the most cost-effective child survival interventions globally at ~\$1–3 per child per year for biannual supplementation. Scaling to near-universal coverage could prevent an estimated 823,000 annual under-5 deaths [PMID 26869575, abstract-only].

Scalability: Proven national. Delivered through national immunisation days, child health weeks, and routine health contacts. Over 80 countries have national VAS programmes.

Government Scaling Pathway: Integrate with existing EPI/immunisation platforms; leverage community health worker networks for biannual delivery. However, as under-5 mortality and VAD prevalence decline, the absolute benefit per child decreases — periodic reassessment of programme necessity is warranted.

2.2 Iron-Folic Acid Supplementation in Pregnancy

Evidence Rating: A

Supported by >20 meta-analyses across the database, including multiple Cochrane reviews. Universal policy recommendation in virtually all LMICs.

Key Effect Sizes:

Outcome	Effect	Source
Maternal anaemia (IFA vs FA alone)	RR 0.52 (95% CI: 0.41–0.66); 5 studies, GRADE moderate	PMID 32075071, fulltext
Iron supplementation → anaemia (children)	RR 0.55 (95% CI: 0.44–0.70)	PMID 31973225, fulltext
LBW (iron vs placebo)	RR 0.88 (95% CI: 0.78–0.99); GRADE high	PMID 32075071, fulltext
Haemoglobin increase (iron in pregnancy)	MD +7.80 g/L (95% CI: 4.08–11.52)	PMID 32075071
Folic acid → mean birthweight	MD +0.37 kg (95% CI: 0.18–0.43); 9 studies	PMID 31680411, fulltext
Folic acid → LBW	OR 0.59 (95% CI: 0.47–0.74)	PMID 31680411

Outcome	Effect	Source
Nutrition education → IFAS compliance	OR 2.80 (95% CI: 2.04–3.83)	PMID 39545365, fulltext
Nutrition education → anaemia prevalence	RR 0.66 (95% CI: 0.54–0.80)	PMID 39545365

Intermittent vs. Daily Supplementation (from Cochrane reviews): - Similar birth outcomes: LBW RR 0.82 (95% CI: 0.55–1.22); birthweight MD +5.13 g (NS) [PMID 26482110] - Fewer side effects with intermittent: RR 0.56 (95% CI: 0.37–0.84) [PMID 26482110] - Better adherence with intermittent: RR 1.60 (95% CI: 1.34–1.91) [PMID 39780191, fulltext] - Daily produces marginally higher Hb: MD +0.24 g/dL (95% CI: 0.12–0.35) [PMID 39780191]

Compliance Challenge: Pooled IFAS compliance in Sub-Saharan Africa is only 39.2% (95% CI: 30.8–48.3%) [PMID 33852614, fulltext]. Key determinants: counselling on IFAS (OR 1.96), knowledge about anaemia (OR 5.42), and ANC visit frequency.

Mechanism: Iron corrects iron deficiency anaemia, supporting oxygen transport and fetal growth. Folic acid prevents neural tube defects and supports DNA synthesis critical for rapid placental and fetal cell division.

Cost-Effectiveness: Very High. IFA tablets cost ~\$0.50–2.00 per pregnancy. Among the lowest cost-per-DALY-averted interventions.

Scalability: Proven national. Part of standard ANC packages in nearly all LMICs. Main bottleneck is compliance/adherence rather than supply.

Government Scaling Pathway: Strengthen ANC platforms with nutrition counselling (which triples compliance); consider intermittent regimens where side effects limit adherence; integrate with mHealth reminders [PMID 30609297, abstract-only — limited evidence but promising].

2.3 Multiple Micronutrient Supplementation in Pregnancy (MMS vs. IFA)

Evidence Rating: A

Three Cochrane reviews (2017, 2019) and multiple IPD meta-analyses consistently show MMS is superior to IFA for birth outcomes.

Key Effect Sizes:

Outcome	Effect	Source
Low birthweight	RR 0.88 (95% CI: 0.85–0.91); 18 trials, n=68,801, high-certainty	PMID 30873598 (Cochrane 2019)
Small-for-gestational-age	RR 0.92 (95% CI: 0.88–0.97); 17 trials, moderate-certainty	PMID 30873598
Very preterm birth	RR 0.81 (95% CI: 0.71–0.93); 4 trials	PMID 30873598
LBW (all studies)	RR 0.85 (95% CI: 0.77–0.93); 28 studies, GRADE high	PMID 32075071, fulltext

Outcome	Effect	Source
Stillbirths (MMS vs IFA)	RR 0.91 (95% CI: 0.86–0.98); 22 studies	PMID 32075071
LBW in anaemic women (Hb <110)	RR 0.81 (95% CI: 0.74–0.89); p-interaction 0.049	PMID 29025632, abstract-only
6-month infant mortality in anaemic women	RR 0.71 (95% CI: 0.60–0.86)	PMID 29025632
Neonatal mortality — female neonates	RR 0.85 (95% CI: 0.75–0.96)	PMID 29025632
UNIMMAP formulation → LBW	RR 0.74 (95% CI: 0.61–0.90); 11 studies	PMID 32075071

No significant effect on: - Perinatal mortality: RR 1.00 (95% CI: 0.90–1.11); high-certainty [PMID 30873598] - Neonatal mortality: RR 1.00 (95% CI: 0.89–1.12); high-certainty [PMID 30873598] - Maternal anaemia vs. IFA: RR 1.04 (95% CI: 0.94–1.15) — MMS does not outperform IFA for anaemia [PMID 30873598]

Important Safety Signal: Sensitivity analysis excluding one Indonesian trial showed significantly increased early neonatal mortality: OR 1.40 (95% CI: 1.08–1.82) [PMID 20120796, fulltext]. This finding has not been replicated in subsequent larger analyses but warrants monitoring.

Effect Modification: - Birthweight gain concentrated in mothers with BMI ≥ 20 : +39 g; nullified in BMI <20: –6 g (interaction $p < 0.001$) [PMID 20120795, fulltext] - Greater benefit with early initiation (<20 weeks) and high adherence ($\geq 95\%$) [PMID 29025632] - Gestational age assessment method does not materially alter findings [PMID 37002655, fulltext]

Mechanism: Addresses multiple concurrent micronutrient deficiencies (iron, folic acid, zinc, vitamins A/B/C/D/E) that limit fetal growth through independent and synergistic pathways.

Cost-Effectiveness: Very High. MMS costs ~\$1.50–3.50 per pregnancy — marginally more than IFA but prevents 12% more LBW births.

Scalability: Proven national. WHO conditionally recommended transition from IFA to MMS in 2020. Several LMICs (Bangladesh, Tanzania, Burkina Faso) have begun national rollout.

Government Scaling Pathway: Replace IFA with MMS in ANC supply chains. Use existing procurement and distribution infrastructure. UNIMMAP formulation standardises composition across manufacturers.

2.4 Large-Scale Food Fortification (Iron, Iodine, Folic Acid)

Evidence Rating: A

One landmark systematic review (PMID 30997493, fulltext) covering 136 studies across three decades provides comprehensive evidence for the impact of mandatory fortification programmes.

Key Effect Sizes:

Nutrient/Vehicle	Outcome	Effect	Source
Iron (flour, sauce)	Anaemia (all ages)	RR 0.66 (95% CI: 0.59–0.74)	PMID 30997493, fulltext

Nutrient/Vehicle	Outcome	Effect	Source
Iron	Anaemia (WRA)	RR 0.66 (95% CI: 0.58–0.76)	PMID 30997493
Iron	Anaemia (pregnant women)	RR 0.73 (95% CI: 0.64–0.84)	PMID 30997493
Iron	Iron deficiency	RR 0.42 (95% CI: 0.32–0.56)	PMID 30997493
Iodine (salt)	Iodine deficiency (school-age)	RR 0.25 (95% CI: 0.21–0.29)	PMID 30997493
Iodine	Goitre prevalence	OR 0.26 (95% CI: 0.16–0.43)	PMID 30997493
Folic acid (flour)	Neural tube defects	OR 0.59 (95% CI: 0.49–0.70)	PMID 30997493
Folic acid	Anencephaly	RR 0.49 (95% CI: 0.40–0.60)	PMID 30997493
Folic acid	Spina bifida	OR 0.66 (95% CI: 0.53–0.82)	PMID 30997493
Folic acid	Folate deficiency (WRA)	RR 0.20 (95% CI: 0.15–0.25)	PMID 30997493
MMN fortification (various foods)	Anaemia	RR 0.68 (95% CI: 0.56–0.84)	PMID 31849042, abstract-only
MMN fortification	Iron deficiency anaemia	RR 0.28 (95% CI: 0.19–0.39)	PMID 31849042
MMN fortification	Vitamin A deficiency	RR 0.42 (95% CI: 0.28–0.62)	PMID 31849042
Double-fortified salt (iron+iodine)	Anaemia	RR 0.59 (95% CI: 0.46–0.77)	PMID 29767699, abstract-only
DFS	IDA	RR 0.37 (95% CI: 0.25–0.54)	PMID 29767699

Mechanism: Adding micronutrients to widely consumed staple foods (flour, salt, oil, sugar, condiments) provides population-wide coverage without requiring individual behaviour change.

Cost-Effectiveness: Very High. Large-scale fortification costs \$0.05–0.50 per person per year — among the highest-return public health investments. Copenhagen Consensus ranks fortification as a top-3 development investment.

Scalability: Proven national. Salt iodisation covers >120 countries. Flour fortification mandated in >80 countries. Infrastructure leverages existing food-processing industry.

Government Scaling Pathway: Establish mandatory fortification standards through food safety regulation; monitor compliance through national quality assurance systems; leverage existing mill/processor infrastructure.

2.5 Exclusive Breastfeeding Promotion (0–6 Months)

Evidence Rating: A

Supported by the landmark Lancet Breastfeeding Series (PMID 26869575, 7,924 citations) and multiple Cochrane reviews.

Key Effect Sizes:

Outcome	Effect	Source
Scaling to near-universal: deaths averted	823,000 under-5 deaths/year preventable	PMID 26869575, abstract-only
Under-5 mortality per 1% EBF increase	3.4 per 1,000 children/year reduction	PMID 33622796, abstract-only
10% EBF improvement → economic savings	~\$1 billion in SSA	PMID 33622796
Undernutrition (breastfeeding vs. not)	OR 0.82 (95% CI: 0.68–0.99)	PMID 35844158, abstract-only
Stunting (breastfeeding vs. not)	OR 0.73 (95% CI: 0.55–0.95)	PMID 35844158
GI infection reduction (6 mo EBF vs. 3–4 mo)	Significant reduction (multiple studies)	PMID 22895934 (Cochrane), abstract-only
Cryptosporidium protection (breastfeeding)	OR 0.40 (95% CI: 0.13–1.22) — suggestive	PMID 29879110, fulltext

EBF Duration: Cochrane review (PMID 22895934, 1,699 citations) confirms no growth deficits with 6-month vs. 3–4-month EBF, with significant reduction in gastrointestinal infections. Iron supplementation needed in settings with suboptimal newborn iron stores.

Mechanism: Breast milk provides complete nutrition, passive immunity (IgA, lactoferrin), and growth factors; EBF reduces pathogen exposure through contaminated food/water.

Cost-Effectiveness: Very High. Breastfeeding promotion through health systems is extremely cost-effective (cost per DALY averted \$1–5 in many settings).

Scalability: Proven national. Baby-Friendly Hospital Initiative implemented in >150 countries. Requires policy support (maternity leave, marketing restrictions) and health worker training.

Government Scaling Pathway: Enforce the International Code of Marketing of Breast-milk Substitutes; expand BFHI; train community health workers in breastfeeding counselling; enact maternity protection legislation.

2.6 Complementary Feeding Education and Food Provision

Evidence Rating: A

Consistently supported across 4 meta-analyses and 2 Cochrane/systematic reviews in the database (PMID 21501443, 24564534, 28904113, 32710657).

Key Effect Sizes:

Outcome	Effect	Source
Education alone — food secure		
HAZ	SMD 0.22 (95% CI: 0.01–0.43)	PMID 24564534, fulltext
Height gain	SMD 0.35 (95% CI: 0.08–0.62)	PMID 24564534
Weight gain	SMD 0.40 (95% CI: 0.02–0.78)	PMID 24564534
LAZ	SMD 0.11 (95% CI: 0.01–0.22)	PMID 28904113, abstract-only
Education alone — food insecure		
HAZ	SMD 0.25 (95% CI: 0.09–0.42)	PMID 24564534, fulltext
WAZ	SMD 0.26 (95% CI: 0.12–0.41)	PMID 24564534
Stunting	RR 0.68 (95% CI: 0.60–0.76)	PMID 24564534
Food provision ± education — food insecure		
HAZ	SMD 0.39 (95% CI: 0.05–0.73)	PMID 24564534, fulltext
WAZ	SMD 0.26 (95% CI: 0.04–0.48)	PMID 24564534
Weight gain (provision)	WMD 0.34 SD (95% CI: 0.11–0.56)	PMID 21501443, fulltext
Linear growth (provision)	WMD 0.26 SD (95% CI: 0.08–0.43)	PMID 21501443
Linear growth (education)	WMD 0.21 SD (95% CI: 0.01–0.41)	PMID 21501443
Family education — WAZ at 12 mo	MD 0.15 SD (95% CI: 0.07–0.22); moderate-certainty	PMID 32710657, abstract-only
Family education — HAZ at 12 mo	MD 0.12 SD (95% CI: 0.05–0.19); moderate-certainty	PMID 32710657

LiST Model Estimates: Complementary food provision reduces stunting by 67% in food-insecure populations; education alone reduces stunting by 30% in food-secure populations (ages 6–24 months) [PMID 24564534].

Mechanism: Education improves feeding practices (meal frequency, dietary diversity, responsive feeding). Food provision directly addresses caloric and micronutrient gaps during the critical 6–24 month window.

Cost-Effectiveness: High. Education-only interventions are low-cost; food provision more resource-intensive but justified in food-insecure settings.

Scalability: Proven subnational. Delivered through ANC/postnatal contacts, community health workers, and IYCF programmes. National programmes exist in India (ICDS), Bangladesh, and Ethiopia.

2.7 Micronutrient Powders (Home Fortification)

Evidence Rating: A

Two Cochrane reviews (2011, 2020 update) with 29 RCTs and 33,147 children provide high-certainty evidence.

Key Effect Sizes:

Outcome	Effect	Source
Anaemia (MNP vs. placebo)	RR 0.82 (95% CI: 0.76–0.90); 16 studies, n=9,927, moderate-certainty	PMID 32107773 (Cochrane 2020), abstract-only
Iron deficiency	RR 0.47 (95% CI: 0.39–0.56); 7 studies, high-certainty	PMID 32107773
Haemoglobin	MD +2.74 g/L (95% CI: 1.95–3.53); 20 studies	PMID 32107773
Ferritin	MD +12.93 µg/L (95% CI: 7.41–18.45)	PMID 32107773
IDA (MNP vs. placebo, 2011)	RR 0.49 (95% CI: 0.35–0.67)	PMID 21901727, abstract-only
MNP vs. daily iron — diarrhoea	RR 0.52 (95% CI: 0.38–0.72) — less diarrhoea with MNP	PMID 32107773
MNP efficacy — anaemia	RR 0.76 (95% CI: 0.69–0.84)	PMID 31973225, fulltext
MNP efficacy — IDA	RR 0.45 (95% CI: 0.34–0.58)	PMID 31973225

Adherence: Pooled high-adherence rate: 63.3% (95% CI: 51.1–74.6%). Strongest predictor of adherence: absence of side effects (OR 2.77, 95% CI: 2.46–3.11). Knowledge of MNPs and social norms also significant [PMID 34658128, fulltext].

Mechanism: Sachets of encapsulated iron + zinc + vitamin A (minimum) sprinkled onto prepared foods at point of use, avoiding palatability issues of liquid iron.

Cost-Effectiveness: High. ~\$3.60 per child per year for 60 sachets.

Scalability: Proven national. National programmes in Bangladesh, Nepal, Kenya, Pakistan. WHO recommends for populations where anaemia prevalence >20%.

2.8 Preventive Zinc Supplementation (Children)

Evidence Rating: A

Two Cochrane reviews (2014, 2023 update) with 80–96 RCTs and >200,000 children.

Key Effect Sizes:

Outcome	Effect	Source
Diarrhoea incidence	RR 0.87 (95% CI: 0.85–0.89); 26 studies, moderate-certainty	PMID 24826920 (Cochrane 2014), abstract-only
All-cause mortality	RR 0.93 (95% CI: 0.84–1.03); 16 studies, high-certainty	PMID 36994923 (Cochrane 2023), abstract-only
LRTI mortality	RR 0.86 (95% CI: 0.64–1.15); moderate-certainty	PMID 36994923
ALRI by specific clinical criteria	IRR 0.65 (95% CI: 0.52–0.82)	PMID 20156999, abstract-only

Outcome	Effect	Source
Pneumonia morbidity	RR 0.81 (95% CI: 0.73–0.90); 6 RCTs	PMID 21501441, fulltext
Diarrhoea morbidity	RR 0.87 (95% CI: 0.81–0.94); 14 trials	PMID 21501441
Height (growth)	SMD –0.09 (95% CI: –0.13 to –0.06); very small positive effect	PMID 24826920

No significant mortality benefit demonstrated at high certainty. Benefits primarily in morbidity reduction. Adverse effect: vomiting (RR 1.29, 95% CI: 1.14–1.46).

Mechanism: Zinc supports immune function, gut epithelial integrity, and cellular growth. Reduces duration and severity of diarrhoeal and respiratory episodes.

Cost-Effectiveness: High. ~\$1–2 per child per year.

Scalability: Proven subnational. Integrated into child health programmes in multiple countries. WHO recommends in areas with high zinc deficiency prevalence.

2.9 Zinc for Diarrhoea Treatment

Evidence Rating: A

Cochrane review (PMID 27996088, fulltext) with 33 RCTs and 10,841 children.

Key Effect Sizes:

Outcome	Effect	Source
Diarrhoea duration (children >6 mo)	MD –11.46 hours (95% CI: –19.72 to –3.19); low certainty	PMID 27996088, fulltext
Diarrhoea persisting to day 7 (>6 mo)	RR 0.73 (95% CI: 0.61–0.88); moderate certainty	PMID 27996088
Duration in malnourished children	MD –26.39 hours (95% CI: –36.54 to –16.23); high certainty	PMID 27996088
Persistent diarrhoea duration	MD –15.84 hours (95% CI: –25.43 to –6.24); moderate certainty	PMID 27996088

Critical subgroup finding: No benefit in children <6 months — duration MD +5.23 hours (95% CI: –4.00 to 14.45) and increased vomiting risk (RR 1.54, 95% CI: 1.05–2.24) [PMID 27996088].

Mechanism: Zinc restores intestinal mucosal integrity and enzymatic function disrupted by diarrhoeal illness, accelerating recovery.

Cost-Effectiveness: Very High. Part of WHO-recommended ORS + zinc treatment protocol. Zinc tablets cost ~\$0.50 per treatment course.

Scalability: Proven national. Included in WHO/UNICEF guidelines for diarrhoea management. Implemented nationally in >40 countries through IMCI protocols.

2.10 Management of Severe Acute Malnutrition (CMAM/RUTF)

Evidence Rating: A

Multiple systematic reviews (PMID 24564235, 31906272) covering SAM treatment protocols.

Key Effect Sizes:

Outcome	Effect	Source
RUTF vs. CSB — SAM recovery	RR 1.51 (95% CI: 1.04–2.20)	PMID 24564235, fulltext
WHO protocol vs. standard care — mortality	RR 0.52 (95% CI: 0.43–0.64) — 48% reduction	PMID 24564235 (citing Lancet 2008)
RUTF weight gain vs. CSB	MD +1.27 g/kg/day (95% CI: 0.16–2.38)	PMID 24564235
RUTF vs. F100 (community) weight gain	MD +5.5 g/kg/day (95% CI: 2.92–8.08)	PMID 31906272, fulltext
Antibiotics + RUTF → mortality	RR 0.74 (95% CI: 0.55–0.98); 3 studies, moderate-certainty	PMID 31906272
Antibiotics + RUTF → recovery	RR 1.06 (95% CI: 1.03–1.08); high-certainty	PMID 31906272
Antibiotics → hospitalisation	RR 0.89 (95% CI: 0.82–0.98); high-certainty	PMID 31906272

Mechanism: RUTF (peanut-based, energy-dense, micronutrient-fortified paste) provides 500 kcal/92 g with complete micronutrient profile in shelf-stable, ready-to-eat format suitable for home-based care.

Cost-Effectiveness: High. CMAM costs ~\$200 per child treated, with >75% recovery rates. Cost per DALY averted estimated at \$42–53 in high-burden settings.

Scalability: Proven subnational. National CMAM programmes in Ethiopia, Niger, Nigeria, Pakistan, and others. Local RUTF production established in several countries and shown equivalent to imported [PMID 24564235].

Government Scaling Pathway: Integrate into primary health care; train community health workers for screening (MUAC) and follow-up; establish local RUTF production. Routine antibiotics (amoxicillin) should be included — they reduce SAM mortality by 26%.

3. Tier 2 Interventions

Strong evidence (A or B+) + scalable with investment

3.1 Lipid-Based Nutrient Supplements for Children (6–23 Months)

Evidence Rating: A

Cochrane/systematic review (PMID 31046132) with 17 RCTs, n=23,200, plus comprehensive analysis in PMID 31973225.

Key Effect Sizes:

Outcome	Effect	Source
Moderate stunting (LNS vs. no intervention)	RR 0.93 (95% CI: 0.88–0.98); 9 studies, moderate quality	PMID 31046132, abstract-only
Severe stunting	RR 0.85 (95% CI: 0.74–0.98); moderate quality	PMID 31046132
Moderate wasting	RR 0.82 (95% CI: 0.74–0.91); moderate quality	PMID 31046132
Moderate underweight	RR 0.85 (95% CI: 0.80–0.91); moderate quality	PMID 31046132
Anaemia	RR 0.79 (95% CI: 0.69–0.90); low quality	PMID 31046132
Stunting (efficacy)	RR 0.90 (95% CI: 0.84–0.96)	PMID 31973225, fulltext
LAZ	MD +0.11 (95% CI: 0.05–0.17)	PMID 31973225
All-cause mortality (LNS vs. non-LNS)	RR 0.73 (95% CI: 0.59–0.89); 13 trials	PMID 31697329, abstract-only
LNS vs. FBF — stunting	RR 0.89 (95% CI: 0.82–0.97); moderate quality	PMID 31046132
LNS vs. MNP — underweight	RR 0.88 (95% CI: 0.78–0.99); moderate quality	PMID 31046132

Mechanism: SQ-LNS provides essential fatty acids, high-quality protein, and micronutrients in a lipid matrix that enhances absorption and provides energy density appropriate for the small gastric capacity of young children.

Cost-Effectiveness: Moderate. ~\$50–60 per child per year for daily SQ-LNS — higher than MNP or supplementation but addresses broader nutrient gaps.

Scalability: Growing. National programmes emerging in Bangladesh, Burkina Faso. Supply chain more complex than MNP; local production being established.

3.2 Prenatal Small-Quantity LNS

Evidence Rating: B+

Individual participant data meta-analysis (PMID 39154665, fulltext) of 4 RCTs (~8,530 women) provides the strongest evidence.

Key Effect Sizes:

Outcome	Effect	Source
Birthweight (SQ-LNS vs. IFA)	MD +49 g	PMID 39154665, fulltext (IPD)
Birth length	MD +0.2 cm	PMID 39154665

Outcome	Effect	Source
LBW risk reduction	11%	PMID 39154665
Newborn stunting reduction	17%	PMID 39154665
Low head circumference z-score	15% reduction	PMID 39154665
Birthweight (LNS vs. IFA, Cochrane)	MD +53.28 g (95% CI: 28.22–78.33)	PMID 30168868, abstract-only
Birth length	MD +0.24 cm (95% CI: 0.11–0.36)	PMID 30168868
SGA	RR 0.94 (95% CI: 0.89–0.99)	PMID 30168868
Newborn stunting	RR 0.82 (95% CI: 0.71–0.94)	PMID 30168868

SQ-LNS vs. MMS: No significant differences for most outcomes — comparable effectiveness [PMID 39154665].

Female infants benefit more: At 6 months, underweight reduced by 26% (95% CI: 7–41%) in female infants [PMID 39154665].

Scalability: Growing. Evidence base still developing. Higher cost than MMS. Trials concentrated in 4 countries.

3.3 Management of Moderate Acute Malnutrition (RUSF/LNS)

Evidence Rating: B+

Outcome	Effect	Source
Recovery (RUSF vs. CSB)	RR 1.07 (95% CI: 1.02–1.13)	PMID 31906272, fulltext
Recovery (LNS vs. FBF)	RR 1.05 (95% CI: 1.01–1.09)	PMID 34535798, abstract-only
Recovery (specially formulated food vs. standard care)	RR 1.29 (95% CI: 1.20–1.38); moderate quality	PMID 23794237, abstract-only
Weight-for-height (vs. standard care)	MD +0.20 z-score (95% CI: 0.03–0.37)	PMID 23794237
Severe wasting prevention (RUSF vs. CSB)	RR 0.74 (95% CI: 0.57–0.95); moderate quality	PMID 31906272

Scalability: Proven subnational. Integrated into supplementary feeding programmes. LNS marginally superior to CSB but both effective.

3.4 Iron Supplementation for School-Age Children and Adolescents

Evidence Rating: B+

Outcome	Effect	Source
Cognitive scores (school-age)	SMD 0.50 (95% CI: 0.11–0.90)	PMID 24130243, abstract-only

Outcome	Effect	Source
IQ in anaemic children	MD +4.55 (95% CI: 0.16–8.94)	PMID 24130243
Haemoglobin (5–24 y, iron vs. placebo)	MD +5.81 g/L (95% CI: 4.19–7.44)	PMID 40063075, abstract-only
Haemoglobin (anaemic populations)	MD +10.24 g/L (95% CI: 5.10–15.39)	PMID 40063075
Serum ferritin (WIFAS in SSA adolescents)	Hedge's g = 0.53 (95% CI: 0.28–0.78)	PMID 38862227, fulltext
Haemoglobin (WIFAS, school setting)	Hedge's g = 0.23 (95% CI: 0.12–0.35)	PMID 38862227

Scalability: Proven subnational. WHO recommends weekly IFA for menstruating adolescents in areas with >20% anaemia prevalence. School-based delivery platforms established in several countries.

3.5 Nutrition Education During Pregnancy

Evidence Rating: B+

Outcome	Effect	Source
IFAS compliance	OR 2.80 (95% CI: 2.04–3.83); 11 studies	PMID 39545365, fulltext
Anaemia prevalence	RR 0.66 (95% CI: 0.54–0.80); 12 studies	PMID 39545365
Haemoglobin	SMD 0.88 (95% CI: 0.63–1.13); 39 studies	PMID 39545365
Birthweight in urban slums	MD +478.44 g (95% CI: 423.55–533.32)	PMID 31204795, abstract-only

Mechanism: Education increases knowledge, self-efficacy, and motivation for supplement adherence and dietary improvement.

Cost-Effectiveness: High. Low-cost intervention delivered through existing ANC platforms.

Scalability: Proven subnational. Can be integrated into routine ANC in all settings.

3.6 Community Case Management of Childhood Pneumonia

Evidence Rating: B+

Outcome	Effect	Source
ALRI mortality (0–5 years)	35% reduction (95% CI: 18–48%)	PMID 20348118, fulltext
ALRI mortality (infants)	42% reduction (95% CI: 33–55%)	PMID 20348118
All-cause mortality (0–5 years)	21% reduction (95% CI: 12–30%)	PMID 20348118

Outcome	Effect	Source
Oxygen therapy → mortality	35% reduction (95% CI: 22–48%)	PMID 20348118

Scalability: Proven subnational. Part of iCCM strategy deployed in >30 countries. Zinc and vitamin A are ineffective as pneumonia adjunct treatments [PMID 20348118].

3.7 WASH Interventions (Combined)

Evidence Rating: B

Outcome	Effect	Source
HAZ (pooled, 10 studies)	SMD 0.14 (95% CI: 0.09–0.19)	PMID 31272479, fulltext
HAZ (children <2 years)	SMD 0.20 (95% CI: 0.11–0.29)	PMID 31272479
HAZ (combined WASH)	SMD 0.15 (95% CI: 0.09–0.20)	PMID 31272479
HAZ (Cochrane, cluster-RCTs only)	MD 0.08 (95% CI: 0.00–0.16) — borderline	PMID 23904195, abstract-only

Key finding: Combined WASH interventions show modest but consistent HAZ improvement. Individual WASH components (latrine only, handwashing only) generally show null effects.

Scalability: Proven subnational. CLTS (Community-Led Total Sanitation) programmes deployed nationally in several countries.

3.8 Conditional Cash Transfers

Evidence Rating: B

Outcome	Effect	Source
Peru CCT (boys, ≥2 years exposure) — HAZ	+0.43 (95% CI: 0.09–0.77)	PMID 31666032, fulltext
Mexico “Oportunidades” (doubled cash) — HAZ	+0.20 (95% CI: 0.09–0.30); stunting –10%	PMID 31666032
Mexico (additional 18 mo early) — height	+1.5 cm (p=0.029)	PMID 31666032
Financial incentives → colostrum receipt	MD +0.22 (95% CI: 0.08–0.35)	PMID 24564520, fulltext

76% of included studies successfully reduced stunting; 65% showed both stunting reduction and economic improvement [PMID 31666032].

Scalability: Proven national. Mexico’s Prospera/Oportunidades, Brazil’s Bolsa Família, and others demonstrate national-scale feasibility.

3.9 IMCI Strategy

Evidence Rating: B

Outcome	Effect	Source
Under-5 mortality	RR 0.85 (95% CI: 0.78–0.93); 2 trials, low certainty	PMID 27378094, fulltext
Infant mortality (India, IMNCI)	HR 0.85 (95% CI: 0.77–0.94)	PMID 27378094
Neonatal mortality (home births, India)	HR 0.80 (95% CI: 0.68–0.93)	PMID 27378094

Little to no effect on stunting (RR 0.94, 95% CI: 0.84–1.06) or wasting [PMID 27378094].

Scalability: Proven subnational. WHO-recommended, implemented in >100 countries.

4. Tier 3 Interventions

Promising evidence (B or C+) + plausible scaling pathway

4.1 Balanced Protein-Energy Supplementation in Pregnancy

Evidence Rating: B

Birthweight improvement: SMD 0.20 (95% CI: 0.03–0.38); 7 RCTs, n=2,367 [PMID 25857334, fulltext]. No significant effect on birth length or head circumference. Effect larger when no control supplement given (greater nutritional deprivation). Evidence base small.

4.2 Calcium Supplementation in Pregnancy

Evidence Rating: B

Pre-eclampsia reduction (calcium alone): RR 0.30 (95% CI: 0.17–0.52); 3 studies [PMID 32075071, fulltext]. No effect on LBW (RR 0.99) or preterm birth. WHO recommends in populations with low calcium intake.

4.3 Vitamin D Supplementation in Pregnancy

Evidence Rating: C+

LBW reduction: RR 0.40 (95% CI: 0.23–0.71); 3 trials — but limited to 5 small trials, mostly high-income [PMID 22742603, fulltext]. Maternal serum 25(OH)D dramatically improved: SMD 1.52 (95% CI: 0.98–2.07) [PMID 40752545, fulltext]. Preterm birth (strictly vitamin D): RR 0.33 (95% CI: 0.17–0.62) in 2 studies [PMID 32075071]. Evidence insufficient for LMIC-specific policy.

4.4 Agricultural Interventions

Evidence Rating: C

Improved dietary diversity in 19/23 studies but no significant effect on anthropometric outcomes or haemoglobin [PMID 22251864, fulltext]. Biofortification and home gardens show promise for vitamin A intake but growth evidence weak.

4.5 Growth Monitoring and Promotion

Evidence Rating: C

Very low-certainty evidence of benefit. No clear effect on WAZ or HAZ across 6 studies [PMID 37823471, fulltext]. Home-based GMP may modestly improve WAZ: MD 0.183 (95% CI: 0.037–0.328) [PMID 37823471].

4.6 Egg-Based Supplementation

Evidence Rating: C+

Emerging evidence from sub-Saharan Africa (7 studies, n=3,673): - WAZ: MD 0.33 (95% CI: 0.11–0.55) [PMID 42137196, fulltext] - WHZ: MD 0.30 (95% CI: 0.12–0.48) [PMID 42137196] - HAZ: MD 0.05 (95% CI: –0.05 to 0.14) — no effect on linear growth [PMID 42137196]

Longer interventions (24 months) show larger effects. Promising food-based approach but evidence still developing.

4.7 Preoperative Oral Nutrition Supplementation (Surgical)

Evidence Rating: B+ (niche population)

Dramatic effects in malnourished GI cancer patients in LMICs: - Surgical complications: RR 0.53 (95% CI: 0.46–0.60); NNT=7 [PMID 35864290, fulltext] - Mortality: RR 0.35 (95% CI: 0.26–0.47) [PMID 35864290]

Evidence primarily from China. Highly relevant for hospital-based nutrition programmes but narrow target population.

5. Cross-Cutting Findings

Pattern 1: The Anaemia Paradox — Iron Deficiency Explains Only a Fraction

Iron deficiency accounts for only **25% of anaemia in preschool children** and **37% in women of reproductive age** — far below the commonly assumed 50% [PMID 27827838, fulltext]. In high-burden settings (>40% anaemia prevalence, rural), iron deficiency explains even less (14–16%). This means iron-focused strategies alone cannot resolve the anaemia burden; inflammation, malaria, helminth infections, and other micronutrient deficiencies must be addressed concurrently.

Pattern 2: The Compliance Bottleneck

Across interventions, efficacy consistently outstrips effectiveness due to poor adherence: - IFAS compliance in SSA: only 39% [PMID 33852614] - MNP adherence: 63% overall, 44% in non-China LMICs [PMID 34658128] - Side effects are the strongest barrier to adherence (OR 2.77 for MNP non-compliance with side effects)

Nutrition education consistently doubles or triples compliance (OR 2.80 for IFAS, PMID 39545365), making it a force-multiplier for any supplementation programme.

Pattern 3: Effect Modification by Nutritional Status at Baseline

Multiple interventions show **greater benefit in more deprived populations**: - MMS: birthweight gain nullified in mothers with BMI <20 but +39 g in BMI ≥20 [PMID 20120795] - MMS: LBW reduction stronger in anaemic women (RR 0.81 vs. 0.88 overall) [PMID 29025632] - Zinc for diarrhoea: MD –26 hours in malnourished children vs. –11 hours overall [PMID 27996088] - VAS: greater mortality benefit in high-VAD populations (RR 0.87 vs. 1.05 in low-VAD) [PMID 30425075]

This argues for **targeted rather than universal** supplementation in settings where deficiency prevalence is declining.

Pattern 4: Nutrition-Specific vs. Nutrition-Sensitive — Complementary, Not Substitutable

The evidence base shows clear limits to nutrition-specific interventions alone: - WASH alone has only borderline HAZ effects (SMD 0.08–0.15) [PMID 31272479, 23904195] - Agricultural interventions improve diet but not anthropometry [PMID 22251864] - CCTs reduce stunting (HAZ +0.20–0.43) through multiple pathways including food, health services, and behaviour change [PMID 31666032] - **76% of studies combining multisectoral policies with nutritional supplementation successfully reduced stunting**, vs. variable results for single-sector approaches [PMID 31666032] - Risk factor analysis confirms the multi-causality: diarrhoea (OR 1.77), low birthweight (OR 2.40), and low maternal education (OR 1.67) are consistent independent predictors of undernutrition [PMID 40886100, fulltext]

Pattern 5: The First 1,000 Days Window Is Real but Not Exclusive

Most interventions show strongest effects during pregnancy and 6–24 months, but the evidence base identifies important gaps: - **Adolescent nutrition**: very limited evidence (10 studies, all Asian LMICs, low quality) [PMID 31878019, fulltext]. Iron may improve Hb (MD 0.42 g/dL) but uncertain effects on anaemia or growth. - **School-age children**: iron supplementation improves cognition (SMD 0.50) and reduces anaemia ~50% [PMID 24130243] - **Elderly malnutrition**: 22.7% prevalence in Ethiopia; low dietary diversity increases odds 4-fold [PMID 37771505, fulltext] - **HIV-positive adults**: nutritional interventions significantly improve BMI (SMD 0.80) and CD4 count (SMD 0.99) [PMID 40460180, fulltext]

Pattern 6: Evidence Quality Varies Dramatically by Intervention

Intervention	Highest GRADE	Number of Cochrane Reviews	Total RCTs in Evidence Base
VAS (child 6–59 mo)	High	3 (2011, 2017, 2022)	47 trials, >1.2M children
MMS in pregnancy	High	2 (2017, 2019)	20 trials, 141,849 women
MNP (home fortification)	High (iron deficiency)	2 (2011, 2020)	29 trials, 33,147 children
Zinc (preventive)	High (mortality)	2 (2014, 2023)	96 trials, 219,584 children

Intervention	Highest GRADE	Number of Cochrane Reviews	Total RCTs in Evidence Base
WASH → nutrition	Low	1 (2013)	5 cluster-RCTs
Adolescent nutrition	Very low	0	10 studies, 10,802
Agricultural interventions	Very low	0	23 studies (mostly non-randomised)

6. Methodology Notes

Evidence Strength Rating Criteria

- **A (Strong):** ≥2 high-quality meta-analyses or Cochrane reviews with consistent findings; high or moderate GRADE certainty; large combined sample sizes (>10,000)
- **B (Moderate):** 1–2 meta-analyses with moderate-quality evidence; some inconsistency across studies or populations; smaller sample sizes
- **B+ (Moderate-Strong):** Between A and B; strong direction of effect but limited by sample size, study number, or GRADE rating
- **C (Emerging):** Limited meta-analytic evidence; low or very low GRADE certainty; small number of trials; inconsistent findings
- **C+ (Emerging-Promising):** Between B and C; suggestive positive direction but insufficient for confident recommendation

Cost-Effectiveness Rating

- **Very High:** <\$100 per DALY averted or established as best-buy by Copenhagen Consensus/DCP3
- **High:** \$100–500 per DALY averted; well-established delivery platforms
- **Moderate:** \$500–1,500 per DALY averted; requires dedicated programme infrastructure
- **Low:** >\$1,500 per DALY averted or insufficient CEA evidence

Scalability Rating

- **Proven national:** Implemented at national scale in ≥3 LMICs with documented coverage
- **Proven subnational:** Successful district/state-level implementation with evidence of feasibility
- **Growing:** Pilot programmes with positive results; scaling pathway identified but not yet proven
- **Requires investment:** Theoretical feasibility but no large-scale implementation evidence

Key Limitations

1. **43 of 100 papers were abstract-only** — effect sizes from these papers could not be independently verified against full-text results sections and may represent selective reporting
2. **Geographic concentration:** Evidence disproportionately from South Asia (India, Bangladesh, Nepal) and East/West Africa. Latin America, Central Asia, and Pacific Island LMICs are underrepresented.
3. **The database contains no dedicated CEA-tier papers** despite the search design including a CEA category — cost-effectiveness ratings therefore rely on external knowledge and are approximate.

4. **Publication bias** was documented in several meta-analyses (IFAS compliance, folic acid/LBW, anaemia interventions) and may inflate effect estimates.
 5. **Heterogeneity** was high ($I^2 > 75\%$) in many pooled analyses, limiting confidence in point estimates.
 6. **One paper (PMID 36321557) is a protocol only** — no results were available for extraction.
 7. **Paper PMID 34100558** (early postnatal discharge) and **PMID 35864290** (preoperative nutrition in cancer surgery) are tangential to the core LMIC nutrition intervention question but were included due to composite scoring.
-

This review synthesises findings from 100 systematically identified papers (93 primary tier, 6 supplementary, 1 nutrition-sensitive). Effect sizes are reported exactly as found in full-text results sections or abstracts. No effect sizes were fabricated. Where full text was not available for a key claim, the paper is marked as abstract-only.